

Subject name	Algorithmic Thinking and Programming with Python	Credits	3Credits
Subject name(English)	Algorithmic Thinking and Programming with Python	Language	English
Contact name	Ian PIUMARTA,KISHIDA Ippei,KURIHARA Masamichi,KATSUMA Yoshiyuki,TOYOKURA Masaki		
Subject number	Students admitted after 2025 : MC222310	授業区分	Face-to-face Course

If the remote class time exceeds half of the total class time, it is a remote course. if it does not, it is a face-to-face course.

Class outline	Information processing is the act of using algorithms to transform data, typically using a computer to perform the algorithm. This course provides students with the necessary skills and knowledge needed to process information and solve problems using algorithms expressed in a general-purpose programming language. Scripting language knowledge and concepts gained by students in previous courses are reviewed and further developed in the context of a popular, interactive, high-level programming language.
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Achievement goals and Grade evaluation standards			
Achievement Goals	Grade evaluation standards		
(3~5 Goals)	Completely achieved	Partly achieved	Requires considerable effort to achieve
On completion of this course participants will ? be able to decompose problems mathematically and algorithmically	Given an abstract problem, it can be expressed clearly and completely using a finite sequence of unambiguous mathematical steps.	A problem can be decomposed into steps but it cannot be expressed clearly, completely, unambiguously, or using a finite number of mathematical steps.	Decomposing algorithms into formal steps cannot be performed.
? understand programming language elements such as values, variables, control flow and abstraction	The concepts, quantities, operations, and structure of a programming language are understood and can be used practically to express an executable version of an algorithm.	Some parts of programming languages remain poorly understood or cannot be applied properly to express an algorithm.	Simple programs for trivial algorithms cannot be expressed properly.
? know how to acquire, manipulate and analyse information using algorithms expressed in a general-purpose programming language	Programming language facilities corresponding to algorithmic inputs, algorithmic manipulations, and consequent display or results are understood and can be used to perform valuable data processing and visualisation functions.	Some kinds of data cannot be properly input or output, or some kinds of data processing remain difficult to accomplish.	Basic input and output, and basic use of data structures and operations, is very difficult or impossible.

Grade evaluation method		
Evaluation method	Percentage	Remarks
In-class interactive tests	50%	Tests cover material presented in previous weeks.
Programming exercises	25%	Practical programming exercises completed in-class after the lecture.
Programming projects	25%	Practical programming projects completed in the final three weeks of the course.

Class schedule and Contents	Remote Class	Teaching method	Extracurricular study	Time required for preparation	Time required for review
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1. Introduction to programming		Lecture and in-class exercises. (Applies to following weeks too.)	Practice programming as much as you possibly can. Think of programs you could write that are related to your interests. Write those programs. (Applies to following weeks too.)	120min	120min
2. Validating your Python interpreter				30min	30min
3. Algorithmic thinking ? concepts				120min	120min
4. Problem solving and algorithmic thinking (1)				30min	30min
5. Algorithmic thinking ? applications				120min	120min
6. Problem solving and algorithmic thinking (2)				30min	30min
7. Programming language basics				120min	120min
8. Representing algorithms as programs				30min	30min
9. Values, constants, and variables				120min	120min
10. Writing and debugging programs (1)				30min	30min
11. Control flow				120min	120min
12. Writing and debugging programs (2)				30min	30min
13. Collections and enumeration				120min	120min
14. Processing collections of data				30min	30min
15. Strings				120min	120min
16. Processing textual information				30min	30min
17. Files				120min	120min
18. Using Files				30min	30min
19. Data structures				120min	12min
20. Using dictionaries to organise information				30min	30min
21. Objects				120min	120min
22. Using abstract data types				30min	30min
23. Abstraction				120min	120min
24. Modeling real-world quantities and concepts as data				30min	30min
25. Acquiring and storing information				120min	120min
26. Information processing (1) data acquisition and storage				30min	30min
27. Processing and visualising information				120min	120min
28. Information processing (2) data visualisation				30min	30min
29. Meta-programming				120min	120min
30. Information processing (3) metaprogramming				30min	30min

Feedback Methods for Assignments (Exams, Reports, etc.)	Tests are online and graded automatically but the University has not provided any mechanism to show each student their individual score confidentially. Students who wish to know their score can request that information from the instructor, provided the demand for that information does not become excessive.
Remote class method	

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Teaching materials(Textbook)				
Title	Author	Publisher	ISBN	Media Type/Remarks

Teaching materials(References)				
Title	Author	Publisher	ISBN	Media Type/Remarks
Think Python	Allen B. Downey	Green Tea Books		A free electronic copy will be provided in-class to all students. https://greenteapress.com/wp/think-python-2e/

Teaching materials(Others)

Course requirements,Rules,academic advice, etc.
Students should be familiar with a scripting system such as the command line shell introduced in Information Literacy. Familiarity with another information processing language, such as MATLAB introduced in Introduction to Numerical Analysis Programming, would be beneficial. Students taking this class should also take Algorithmic Thinking and Programming with Python Exercises. Students should bring a laptop computer to class, with a working Python 3 installation.

Diploma Policies		
	Department of Mechanical and Electrical Systems Engineering Diploma Policy	Completion rate
DP1.1	Graduates shall acquire a specific core body of knowledge, relating it to knowledge in other fields, and use it to solve problems in a changing global society.	10%
DP2.1	Graduates shall be able to collect and use necessary information and data using appropriate methods.	35%
DP2.2	Graduates shall be able to communicate with others using a variety of languages.	15%
DP3.1	Graduates shall be able to logically construct and express their own ideas through multifaceted thinking by utilizing acquired knowledge, skills, and experience.	25%
DP3.2	Graduates shall be able to critically examine a topic of their own choosing, while objectively analyzing any information collected.	0%
DP4.1	Graduates shall demonstrate an ongoing interest in the problems of a changing global society and be able to act proactively and persistently to solve those problems.	0%
DP4.2	Graduates shall be able to act as an autonomous member of society while collaborating with others from diverse backgrounds	15%